1. One-Tailed Z-Test (Right Tail)

Example 1: Mean (σ known)

A machine fills AI-training data disks. The fill amount is **normally distributed** with

- $\sigma = 4$ GB
- Historically mean = 50 GB

A new technician claims the **average fill is now more than 50 GB**.
A sample of $n = 40$ disks has mean = 51.2 GB.
Test the claim at $\alpha = 0.05$.

Solution

Z-statistic:

Critical value (right-tail 0.05):
Since $1.90 > 1.645$,
Reject H .
⇒ Mean is significantly greater than 50 GB.

2. Two-Tailed Z-Test (Mean, σ Known)

Example 2

For a cloud server, response time has $\sigma = 2.5$ ms
A researcher claims the mean response time is **not equal** to 100 ms.
A sample of 50 observations has 2. Use $\alpha = 0.05$.

Solution

Critical values:

Since −2.26 < −1.96, **Reject H**
⇒ Mean is significantly different from 100 ms.

3. One-Tailed t-Test (Left Tail)

Example 3: σ unknown, small n

An AI start-up claims their new optimizer reduces training time to **less than 10 hours**.
A sample of **n = 9** training runs gives: . Test at α = 0.05.

Solution

Critical value (left-tail, 0.05):
Since **−2.25 < −1.86**,
Reject H
⇒ Mean training time is significantly less than 10 hours.

4. Two-Tailed t-Test (Mean, σ unknown)

Example 4

You want to check if the average number of epochs for convergence is **different** from 20.
A sample of n = 12 gives: . Test at α = 0.05.

Solution

Critical value (two-tailed 0.05):
]Since **−1.71** is *not* beyond ±2.201,
Fail to reject H
⇒ No significant evidence that mean ≠ 20.